

```

1  import java.util.Random;
2  import java.util.Scanner;
3
4  public class CovidTestHelper {
5      public static void main(String[] args) {
6          Queue<CovidTest> q = queue5Random();
7          System.out.println(q.toString());
8
9          System.out.println("most sick city " + mostSick(q));
10     }
11
12     // קוד לייצר רשימה רנדומלית
13     public static Queue<CovidTest> queue5Random(){
14         Queue<CovidTest> queue = new Queue<CovidTest>();
15         Random random = new Random();
16         Scanner input = new Scanner(System.in);
17         String name = "";
18         String id = "";
19         boolean sick = false;
20         for(int i = 0; i < 10; i++){
21             queue.insert(new CovidTest(name, id,
22                 random.nextInt(1, 10),
23                 random.nextBoolean()));
24         }
25         return queue;
26     }
27
28     public static int mostSick(Queue<CovidTest> q){
29         Queue<CovidTest> temp = new Queue<CovidTest>();
30         int maxCount = -1;
31         int bigCity = -1;
32         int currentCity = -1;
33         while(!q.isEmpty()){
34             currentCity = sickInCity(q, q.head().getCityCode());
35             if(currentCity > maxCount) {
36                 maxCount = currentCity;
37                 bigCity = q.head().getCityCode();
38             }
39             temp.insert(q.remove());
40         }
41         while(!temp.isEmpty()){
42             q.insert(temp.remove());
43         }
44         return bigCity;
45     }
46
47     public static int sickInCity(Queue<CovidTest> q, int cityCode){
48         Queue<CovidTest> temp = new Queue<CovidTest>();
49         int count = 0;
50         while (!q.isEmpty()){
51             if(q.head().getCityCode() == cityCode && q.head().isSick()){
52                 count++;
53             }

```

```
54     temp.insert(q.remove());
55 }
56 while(!temp.isEmpty()){
57     q.insert(temp.remove());
58 }
59 return count;
60 }
61 }
62 /*
63 [{, , 2, true}, {, , 1, true}, {, , 5, true}, {, , 7, false}, {, , 2, true}, {, , 2, true}, {, , 4, true
64 }, {, , 7, true}, {, , 8, true}, {, , 6, true}]
65 most sick city 2
66
67 * */
```